Hayeon Yun

Evanston, IL | 704-490-1993 | hayeonyun2026@u.northwestern.edu | linkedin.com/in/yun-hayeon/

EDUCATION

Northwestern University — Evanston, IL
B.S. in Biomedical Engineering — Expected Dec 2026
GPA: 3.89 / 4.00

Davidson College — Davidson, NC
Coursework in Biology and Computer Science — Aug 2022 – May 2023
GPA: 4.00 / 4.00

RESEARCH INTERESTS

Design and fabrication of soft, biointegrated electronic systems for sensing, stimulation, and biomedical applications, with interests in tissue-integrated interfaces, wearable and implantable devices, and regenerative medicine.

RESEARCH EXPERIENCE

Querrey Simpson Institute for Bioelectronics, Northwestern University — Evanston, IL
Research Assistant, Organoid Systems Team (PI: Prof. John A. Rogers) — Sep 2025 – Present

- Contribute to the development, fabrication, and characterization of flexible 3D bioelectronic interfaces across engineered heart tissues, living myocardial slices, neuromuscular junction and neurocardiac organoids, neurosheets, and electrical impedance tomography platforms, as well as organoid oxygenation systems
- Developed the compressive-buckling protocol for a dual-cradle neurocardiac interface by experimentally refining device prestrain based on simulated geometry, cradle dimensions, and inter-cradle spacing; established the resulting stretch ratio as the standard fabrication protocol
- Optimized fabrication of a PDMS air-channel system for enhanced organoid oxygenation by refining PVA dissolution, laser micromachining, and 3D assembly, reducing fabrication time from 3–4 days to 1 day and increasing fabrication yield from below 50% to approximately 70%
- Developed a fabrication-tracking application to standardize process documentation, capture stage-specific fabrication and QC data, and support systematic investigation of batch-to-batch yield variability
- Trained and mentored six undergraduate researchers and provided hands-on fabrication instruction to graduate and postdoctoral researchers in microfabrication, Pt-black electroplating, device testing, and process troubleshooting

Laboratory for Organic & Hybrid Bioelectronics, Northwestern University — Evanston, IL
Research Assistant, Aptamer Sensor Team (PI: Prof. Jonathan Rivnay) — Jul 2026 – Present

- Fabricate and characterize aptamer-based electrochemical sensors by varying hydrogel coating conditions and evaluating resulting sensor performance through electrochemical testing

Chemistry for Life Processes Institute, Northwestern University — Evanston, IL
Research Assistant, Tau Protein Team (PI: Prof. Songi Han) — Jan 2025 – Jun 2025

- Performed tau aggregation studies using Thioflavin T fluorescence, gel electrophoresis, and negative-stain TEM to evaluate fibril formation and morphology under varied protein and salt conditions

Korea Advanced Institute of Science and Technology — Daejeon, South Korea
Research Intern, Algorithm Development Team (PI: Prof. Jinkuk Kim) — Jun 2024 – Aug 2024

- Developed a Python pipeline integrating PolyA_DB 3 polyadenylation-site annotations with GTEx-based SpliceAI data, then modified and trained SpliceAI to predict polyadenylation sites alongside splice donor and acceptor sites
- Investigated SeqProp for sequence optimization and designed a strategy to adapt its differentiable sequence-design framework from APARENT to SpliceAI using donor- and acceptor-specific objectives

CONFERENCE PRESENTATIONS & ABSTRACTS

(1) **Yun, H.**, Kang, Y.-W., et al. “Flexible, 3D Electronic Framework for Electromechanical Monitoring and Stretching of Engineered Heart Tissues.” BMES Annual Meeting, 2026. Abstract accepted.
(2) **Yun, H.**, Sama, V., et al. “Flexible, 3D Electronic Framework for Electromechanical Monitoring and Stretching of Engineered Heart Tissues.” CBEC Conference, 2026. Poster.
(3) **Yun, H.**, Kang, Y.-W., et al. “Flexible, 3D Electronic Framework for Electromechanical Monitoring in Engineered Heart Tissues.” 39th U.S.–Korea Conference on Science, Technology, and Entrepreneurship, 2026. Poster.
(4) **Yun, H.**, Kang, Y.-W., et al. “Electrical Signal Recording Using Flexible 3D Bioelectronic Interfaces for Neuromuscular Organ-On-Chip Platforms.” 7th KSEA Midwest Regional Conference, 2026. Poster.

SELECTED MANUSCRIPTS IN PREPARATION

(1) Kang, Y.-W., Chychula, I. A., Kim, J., Li, X., Maeng, W.-Y., Jo, M.-S., Aydin, O., **Yun, H.**, Sama, V., Gauthier, S., Lee, H., Asseily, C., Sachdeva, R., Ding, L., Fox, R., Lee, J., McNally, E. M., Huang, Y., Fullenkamp, D. E. & Rogers, J. A. "A mechanically interactive bioelectronic interface for programmable stimulation, recording and mechanical conditioning of engineered heart tissues."

(2) Kang, Y.-W., Aydin, O., Lyu, Z., Maeng, W.-Y., Lee, J., **Yun, H.**, Sama, V., Chan, K. A., Jo, M.-S., Casler, J. C., Kim, T., Gauthier, S., Ding, L., Lee, H., Wang, Q., Bae, J., Huang, Y., Franz, C. K. & Rogers, J. A. "A cylindrical air-channel array for enhanced oxygenation of three-dimensional organoid cultures."

(3) Kang, Y.-W., Aydin, O., Berger, D., Amato, F., Mossuto, S., Li, X., Jo, M.-S., **Yun, H.**, Sama, V., Gauthier, S., Lee, J., Lee, H., Asseily, C., Sachdeva, R., Ding, L., Fox, R., Huang, Y., Osaki, T., Rogers, J. A. & Vahabikashi, A. "A three-dimensional bioelectronic interface for multimodal interrogation of neuromuscular junction organoids."

HONORS AND AWARDS

Kidney Technology Development Prize, NIH/VentureWell DEBUT Challenge, 2026

- Received a $15,000 team prize from the National Institute of Diabetes and Digestive and Kidney Diseases for UroSense, an automatic multiparametric urinalysis system for continuous hydration assessment in intensive care units

Poster Award, 39th U.S.–Korea Conference on Science, Technology, and Entrepreneurship, 2026

BME Summer Undergraduate Research Award, Northwestern University, 2026

- Awarded $5,400 to support nine weeks of full-time undergraduate research in a Northwestern BME laboratory

Dean's High Honors, 4 quarters, 2024–2026 | **Dean's Honors**, 4 quarters, 2023–2025

SELECTED ENGINEERING & DESIGN PROJECT

Hollister Incorporated | UroSense: Automatic Hydration Assessment in the ICU Evanston, IL

Biomedical Engineering Design Project Sep 2025 – Jun 2026

- Designed and iterated CAD prototypes of a catheter-integrated hydration monitoring system combining a self-emptying siphon chamber, dual reflective IR sensors for urine output measurement, and an integrated refractometer for urine specific gravity
- Designed and executed benchtop validation experiments for the sampling and optical sensing subsystems, demonstrating a mean siphon dump volume of 5.46 ± 0.11 mL across 10 trials and $R^2 > 0.99$ between sensor timing and flow rate
- Integrated and characterized the refractometry subsystem across physiologically relevant artificial urine concentrations, achieving a maximum measurement standard deviation of 0.0002 and a mean urine specific gravity difference of 0.0018 relative to a reference refractometer

INDUSTRY EXPERIENCE

Samsung Biologics | Manufacturing Science & Tech Department Songdo, South Korea

Biomanufacturing Intern, Downstream Processing Jun 2025 – Aug 2025

- Identified incomplete cleaning in the upper corners of downstream chromatography columns and proposed a backflow cleaning strategy to improve fluid coverage of poorly cleaned regions
- Synthesized buffer-mixing data from SOPs, BMRs, MBRs, CAPAs, and deviation reports across 77 historical batches into an Excel reference, consolidating key process parameters such as buffer composition, mixing time, and vessel size

TEACHING AND MENTORING

Northwestern University | Academic Support and Learning Advancement Evanston, IL

Math Peer Tutor Sep 2025 – Mar 2026

- Tutored 15+ students for 4 hours per week across 13 introductory mathematics courses, including multivariable calculus and linear algebra

TECHNICAL SKILLS

Microfabrication & Device Assembly: Photolithography (SU-8, nLOF, S1813), mask alignment, spin coating, Thermal and electron-beam evaporation of Au/Cu, lift-off, wet/dry etching, PDMS casting, oxygen-plasma treatment and bonding, laser micromachining (LPKF), compressive buckling, 3D assembly, Pt-black electroplating

Bioelectronic & Electrochemical Characterization: Electrode impedance measurement, electrophysiological recording (Intan), electrochemical sensor testing, strain-gauge sensing, magnetic actuation, multichannel device QC, optical microscopy

Biological Methods: Protein and buffer preparation, Thioflavin T (ThT) aggregation assays, gel electrophoresis, negative-stain TEM grid preparation and imaging

Programming & Engineering Design: Python, MATLAB, R, ImageJ, SolidWorks, Onshape